

Problem 4.3

In how many years will \$500 accumulate to \$630 at 7.8% annual simple interest?

Solution

For simple interest,

$$A = P(1 + it).$$

Substituting the given values,

$$630 = 500 \left(1 + \frac{7.8}{100} t \right).$$

Dividing by 500,

$$\frac{630}{500} = 1 + \frac{7.8}{100} t.$$

Solving for t ,

$$t = \left(\frac{630}{500} - 1 \right) \frac{100}{7.8}.$$

Therefore,

$$t \approx 3.33 \text{ years.}$$